EXERCISE 9

A Second Solution of an Euler Equation by Reduction of Order

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Reduction of Order • Euler Equation • Wronskian

STATEMENT

> For $t > 0$ the function $y_1(t) = 1/t$ is a solution of $t^2\ddot{y} - 2t\dot{y} - 4y = 0$. Using the previous exercise, find a second, linearly independent solution.

SETUP AND HYPOTHESES

Domain $t > 0$ (so $y_1 = 1/t \neq 0$). The equation $t^2\ddot{y} - 2t\dot{y} - 4y = 0$ is homogeneous with variable coefficients; dividing by t^2 gives the standard form $\ddot{y} + p\dot{y} + qy = 0$ with $p(t) = -\dfrac{2}{t}$, $q(t) = -\dfrac{4}{t^2}$, continuous for $t > 0$. First check that y_1 is a solution:

$$y_1 = t^{-1},\ \ \dot{y}_1 = -t^{-2},\ \ \ddot{y}_1 = 2t^{-3} \ \Rightarrow\ t^2(2t^{-3}) - 2t(-t^{-2}) - 4t^{-1} = 2t^{-1} + 2t^{-1} - 4t^{-1} = 0\ \checkmark$$

FORMULATION

Apply Exercise 8 with $g = 0$: write $y = v\,y_1 = v/t$, so that v must satisfy $y_1\ddot{v} + [2\dot{y}_1 + py_1]\dot{v} = 0$. A second independent solution is $y_2 = v\,y_1$ with v non-constant.

METHOD OF THE CLASS NOTES [NOTES, P. 7]

Reduction of order as in the notes: with the known solution $u = y_1 = 1/t$, write $y = v\,u$; the bracket $u'' + pu' + qu$ vanishes and $z = v'$ satisfies $uz' + (2u' + pu)z = 0$, a first-order equation. The notes stop there; it is separable, and integrating it gives the second solution below.

DERIVATION

The equation for v

Compute the coefficient: $2\dot{y}_1 + p y_1 = 2(-t^{-2}) + \left(-\dfrac{2}{t}\right)\left(\dfrac{1}{t}\right) = -2t^{-2} - 2t^{-2} = -4t^{-2}$. The equation of Exercise 8 is

$$\frac{1}{t}\,\ddot{v} - \frac{4}{t^2}\,\dot{v} = 0 \quad\Longleftrightarrow\quad \ddot{v} = \frac{4}{t}\,\dot{v} \quad \text{(multiplying by } t\text{)}$$

Solve for $w = \dot{v}$

$\dot{w} = \dfrac{4}{t} w$ is separable (or use the integrating factor $e^{-4\ln t} = t^{-4}$): $\dfrac{d}{dt}(t^{-4}w) = t^{-4}\dot{w} - 4t^{-5}w = t^{-4}\left(\dot{w} - \dfrac{4}{t}w\right) = 0$, so $t^{-4}w = c$ and

$$w = \dot{v} = c\,t^4$$

Integrate and multiply by y_1

$$v = \int c t^4\,dt = \frac{c}{5}t^5 + b \quad\Longrightarrow\quad y = \frac{v}{t} = \frac{c}{5}t^4 + \frac{b}{t}$$

The term in b is the known solution $y_1 = 1/t$; the new, independent solution is (taking $c = 5, b = 0$)

$$\boxed{y_2(t) = t^4}$$

Verification

Substitute: $y_2 = t^4$, $\dot{y}_2 = 4t^3$, $\ddot{y}_2 = 12t^2$, so $t^2(12t^2) - 2t(4t^3) - 4t^4 = (12 - 8 - 4)t^4 = 0$ ✓. Wronskian:

$$W[y_1, y_2] = y_1\dot{y}_2 - y_2\dot{y}_1 = \frac{1}{t}(4t^3) - t^4\left(-\frac{1}{t^2}\right) = 4t^2 + t^2 = 5t^2 \neq 0 \quad (t > 0)$$

so y_1 and y_2 are linearly independent on $t > 0$. Abel's identity gives the same dependence on t: $W = W_0 \exp\left(-\int p\,dt\right) = W_0 \exp\left(\int \frac{2}{t} dt\right) = W_0 t^2$, consistent with $5t^2$ ($W_0 = 5$).

INTERPRETATION AND CHECKS

RESULT

A second independent solution is $y_2 = t^4$, and the general solution on $t > 0$ is $y = c_1 t^{-1} + c_2 t^4$.

- Euler cross-check: for $t^2\ddot{y} - 2t\dot{y} - 4y = 0$ the trial $y = t^k$ gives $k(k-1) - 2k - 4 = k^2 - 3k - 4 = (k-4)(k+1) = 0$, so $k = 4, -1$: the same two solutions, found by a different method.
- Dimensional/scaling check: under $t \to \lambda t$ the Euler equation is invariant and the solutions are homogeneous power functions of degree -1 and 4.
- The general solution is not defined at $t = 0$ (where y_1 blows up and the equation is singular), consistent with the restriction $t > 0$.